Research article

Temporal Change in Modeled Connectivity Among Southern African Cheetah (*Acinonyx jubatus jubatus*) Range Cores, 2012–2024

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Abstract

[TODO: ABSTRACT CHECKLIST, SET UP FOR BIODIVERSITY AND CONSERVATION. Your draft below has the right results. The journal wants ONE plain paragraph of 150 to 250 words (your draft is about 290), with no undefined abbreviations and no citations. Citations count as "unspecified references," so every `\parencite` in the draft has to go; name things in words instead (for example "the 2010–2016 southern African range assessment"). One abstract serves all three layouts, so this also works for the capstone.

A GOOD ORDER, one or two sentences each: the problem (range mostly outside parks, connections between range areas matter); the gap (fix 3); what you did, including the fixed cores (fixes 1 and 5); what you found (the 253-pair change, the protection numbers, the 32/13 split; fixes 4 and 6); the tracking check (fix 7); and what it means for conservation, which is the journal's core interest: change was mixed rather than a broad decline, and the connecting land outside parks matters.

FIXES, in order of importance:
1. The fixed cores are missing. Your protocol requires them in the abstract, ideally sentence two: the cores come from 2010–2016 records and were held still, so every change you report is the landscape changing, not cheetahs moving.
2. "current extant range" uses a banned phrase. Say "mapped range" or "2010–2016 range."
3. "very little predictive modeling" describes a forecasting study, but yours looks backward at 2012–2024. Say that past change in connectivity between known range areas has not been measured.
4. "the landscape that is most important for cheetah movement" claims real movement. Say "the land with the highest modeled current."
5. "a multivariate analysis" is vague. Say what it is: a resistance surface built from human pressure, vegetation, and slope, run through least-cost paths and circuit theory.
6. The 161/92 split is out of 253 pairs, while the 32/13 split is out of 45 links. A reader will not know these are different sets; one clause each fixes it.
7. Add the plausibility check in one sentence. The stricter step check is the one to cite: in tracking data from nine free-roaming cheetahs, each day's move tended to end on lower modeled resistance than moves of the same length the animal could have made instead (seven of nine animals; a modest lean, mostly from roads). Do not mention current here; it did not hold up.
8. The opening has grammar slips ("The southern African Cheetah ... has had their range"; "decreased drastically in the past decades" twice). Fix these on your final pass.
9. The KAZA and Iran sentences are background that will not fit in 250 words. Move them to the Introduction if they are not already there.
10. KEYWORDS: the journal wants 4 to 6 keywords that are NOT already in the title. Three of yours repeat it: "landscape connectivity" (title says connectivity), "*Acinonyx jubatus*" (in the title), and "southern Africa" (title says Southern African). Keep "circuit theory," "protected area coverage," and "monitoring prioritisation," and add one to three new ones, for example "least-cost path," "resistance surface," "land-use change," or "Namibia; Botswana" to carry the location.]

The southern African Cheetah (*Acinonyx jubatus jubatus*), has had their range decreased drastically in the past decades. This has only been exacerbated in the past decades due to both natural and anthropogenic reasons, leaving their current extant range majority outside protected areas (Weise et al. 2017b). This leaves barriers like ranchers, roads and other human development in areas where cheetahs will be roaming (Marker et al. 2008). Models looking at habitat suitability have been done in the past, but very little predictive modeling of the movement network permeability. One 2026 KAZA (Kavango-Zambezi Transfrontier Conservation Area) multi-species connectivity study excluded the cheetah outright (Sousa et al. 2026). A similar study was completed for the Asiatic cheetah (*Acinonyx jubatus venaticus*), identifying potential connectivity through the Iranian landscape (Moqanaki and Cushman 2017). This study attempts to identify core areas within southern Africa using population data from the Dryad dataset accompanying Weise et al. (2017b) (Weise et al. 2017a), model connectivity between them using a multivariate analysis, then see how that connectivity has changed between 2012 and 2024. The modeled connection cost changed only slightly overall. Out of the 253 core pairs, 161 became somewhat easier and 92 became harder. Outside the core areas, only about 23% of the landscape that is most important for cheetah movement was protected. This is lower than the 31.26% of protected land across the wider study area. Protection was higher only in the very most connected places. Of the 45 links selected by the model, 32 were kept as the main priority links. The other 13 are still possible connections, but need more field data before they can be treated as priorities.

Keywords: landscape connectivity; circuit theory; *Acinonyx jubatus*; protected area coverage; monitoring prioritisation; southern Africa

1 Introduction

The southern African cheetah has had their range substantially reduced, with their current range estimated to be on majority unprotected land. Cheetah range on protected land is governed by centralized park-management initiatives (Buk et al. 2018). Unprotected land poses a different type of challenge, with landowners, local communities and governmental entities involved. This paper looks at landscape connectivity change between 2012 and 2024. The core range of cheetahs was fixed at the 2010-2016 baseline, so modeled changes in permeability reflected landscape changes rather than changes in the core range itself.

Previous research in the region has predominantly focused on habitat suitability, a topic that is already comprehensively addressed in the existing literature. The objective of my study was not to develop an additional suitability model, but rather to identify and evaluate candidate spatial "pinch points" (i.e., constricted linkages) among core areas. The study most comparable to the present work is a recent multi-species connectivity assessment; however, that analysis explicitly excluded cheetah due to insufficient data. Data limitations were therefore explicitly considered and mitigated to the extent possible in this study, and they provide both the primary justification for the existence of this paper and its main constraint.

Two primary connectivity approaches are commonly used to model these kinds of frameworks: least-cost connectivity and circuit-theoretic connectivity (Adriaensen et al. 2003; McRae et al. 2008). The least-cost approach links core areas by selecting the route with the lowest cumulative "cost," meaning the path of least resistance between cores, based on weight values assigned to each multivariate resistor. In contrast, the circuit-theoretic approach represents the landscape as a resistor network, sends a "current" through it, and highlights redundancy across the broader landscape, relying on the same multivariate resistors. A key limitation of both methods is that you must first assign a resistance "score" or "weight" to every cell (1 km in my case). Zeller et al. (2012) demonstrated that even reasonable alternative assumptions about resistance values can influence results more strongly than the choice of connectivity method itself. For that reason, rather than treating one resistance surface as definitive, I varied the weights along two axes. The first changed the balance among the anthropogenic pressures, comparing my primary weighting against an equal weighting and an infrastructure-emphasis weighting. The second changed how much the surface leaned on vegetation relative to anthropogenic pressure and terrain, at 10, 20 and 40 percent. The vegetation-balanced surface combined with the documented fence treatment carries the primary results, and the remaining combinations are reported as sensitivity rather than as alternative answers. A link is only described as stable here if it survives them. This helps to minimize bias from the selection of resistor weights.

Temporal-change hypothesis
Temporal change in connectivity is expected to be strongest among a minority of links, and the links that are most important to the network are expected to experience the greatest loss between 2012 and 2024. The importance of a link is measured by its "betweenness" in the network, which is the number of shortest paths that end up passing through it. A link with a high betweenness is therefore more important to the network than a link with a low betweenness. This was compared between the 45 selected links.

Protected-area-coverage hypothesis
High-value connectivity ares are those with the highest connectivity values. Cells that are part of the protected areas are expected to have higher connectivity values than cells in the unprotected areas. The proportion of the cheetah's range that is currently protected is expected to be a significant factor in maintaining connectivity. The protected-area used was the most recent available data and was not reconstructed at each historical date. The expectation is that the highest value connectivity areas are located disproportionately outside formal protection areas. This would be refuted if the share of unprotected high current area is not better than the comparison.

Link-stability hypothesis
The stability of a link is defined as its ability to maintain connectivity under varying conditions. Some links may be more sensitive to changes in the landscape due to their location or the surrounding land use (i.e., a link that is surrounded by agricultural land may be more sensitive to changes in land use than a link that is surrounded by protected areas). The expectation is that there will be a small group of links that are most important under all resistance scenarios, while other links will be sensitive to the model changes. This would be refuted if the sensitivity tests add nothing, or they are mostly low.

Monitoring priority
The active monitoring of cheetahs in the wild is as challenging as it is important. The areas in which to focus this monitoring are paramount into making sure that money flowing into conservation is utilized in the most efficient way possible. The highest priority places to monitor is where network importance overlaps with old or sparse cheetah records and fast landscape change.

Throughout, potential structural connectivity refers to modelled landscape permeability between fixed range cores. It is not evidence of cheetah movement, dispersal,

or gene flow. Where a location is described as a pinch point, this means a location of concentrated modelled current or a link-level bottleneck, not an observed crossing.

2 Methods

2.1 Study area and analysis extent

The study area was defined as the southern African cheetah range (Weise et al. 2017b) plus a buffered analysis extent. All layers were aligned to a fixed equal-area projection and a 1 km snap grid, so every raster was directly comparable cell for cell. Cores were derived from the historical confirmed-range baseline and held constant across 2012, 2016, 2020, and 2024. The final network consisted of 23 cores and 45 selected neighboring core pairs (Fig. 1). I used a 1-km working resolution because it represented a compromise between the rougher 10-km occurrence baseline and the finer environmental covariates, which ranged from 30 m to 1 km. This resolution was fine enough to identify meaningful points in the landscape while avoiding the implication of a level of accuracy that the occurrence evidence alone could not provide.

2.2 Range cores

I defined a core area as a contiguous patch of cells that were confirmed presence, with a density of ≥ 0.51 cheetahs per 100 km^2, which was from the density raster's own natural breaks that represent the core areas best (Weise et al. 2017a). The final network consisted of 23 cores, which were held constant across all four temporal snapshots to avoid inventing core movement in the absence of comparable range reassessment. The minimum for the core size was 500 km^2 based on an average of observed ranges per cheetah (Melzheimer et al. 2020).

I represented these 23 core areas as fixed nodes across all four temporal snapshots (2012, 2016, 2020 and 2024). I took each core area and calculated the Euclidean distance to the three nearest neighbors and a minimum spanning tree was added to ensure all cores were connected. This yielded 45 straight-line links. These straight-line links defined the lines used to get the least cost paths based on the added resistance surface rasters (2.6).

2.3 Covariates and their temporal treatment

The predicted temporal resistance values came from the continuous rasters (human development from GHSL, nighttime lights from VIIRS, and VCF (vegetation) from MODIS). Categorical land cover was used for masking and description only, because annual categorical features can switch classes between years for reasons unrelated to overall landscape change.

I also used the static covariates (roads, livestock, terrain, hydrography, fences, protected areas, and core definitions). The static covariates did not contribute *temporal* signal to the resistance surface; however, they did contribute *spatial* signal to the resistance surface. The static covariates were used to describe the landscape and to provide context for the dynamic covariates.

[TODO: TABLE (moved here from Results, 6 October 2026): mention the covariates table in this subsection, e.g. "(Table~`\ref{tab:covariates}`)" after you first list the covariates. It is now the first table in the paper. The gaps in the table itself are listed in the note under it.]

2.4 Resistance scenarios

Resistance values were assigned to represent the relative difficulty of movement through each landscape feature. The terrain component was derived from slope, with steeper slopes assigned higher resistance values. The final balanced map had a weight of 60% for human pressure, 20% for vegetation, and 20% for terrain. Within human pressure, built-up land, roads, and livestock each received 30%, while nighttime lights received 10%. Available fence data was incorporated into the resistance model, with fence cells increasing the combined resistance by a multiplier of 1 if there is no fence, or 25 if there is a fence cell. Fences did not incur a full block to cheetah movement as they are not complete barriers (Cozzi et al. 2013). I gave slopes 10 degrees or less a resistance of 1, and slopes between 10 and 30 degrees a resistance that increased in a straight line to 10. Steeper slopes were assigned a resistance of 10, with the calculated slope being an average within each 1-km cell. This had to be done carefully however, given cheetahs do have an affinity for some sloped terrains (Moqanaki and Cushman 2017). I compared alternative vegetation weights because vegetation cover can influence cheetah habitat use (Verschueren et al. 2024). Different weighting schemes allowed me to test how strongly the modeled resistance surfaces depended on the vegetation weight (Zeller et al. 2012). The final resistance surface was used to model least-cost paths and Circuitscape current flow between core areas. The overall number decisions about the resistance surfaces were made by me, based on the literature cited and were not fitted to any cheetah movement data. Overall, three human-pressure weightings (the main surface, equal, and infrastructure-emphasis), three vegetation weights (10, 20, and 40 percent), and five fence treatments (no fences, documented fences, near-barrier, Kruger documented, and Kruger conservative) were tested. The main surface was the vegetation-balanced surface with documented fences.

2.5 Connectivity modelling

For each core, I ran a least-cost analysis in ArcGIS, which calculated the accumulated cost of moving through every reachable cell. I then used the lowest accumulated cost to draw paths from the starting core to each connected destination. Reusing the accumulated-cost calculation for several destinations produced the same set of pairwise paths with fewer repeat calculations. I also used Circuitscape 5.17.1 to model the distribution of possible movement routes between each pair of cores. The anal-

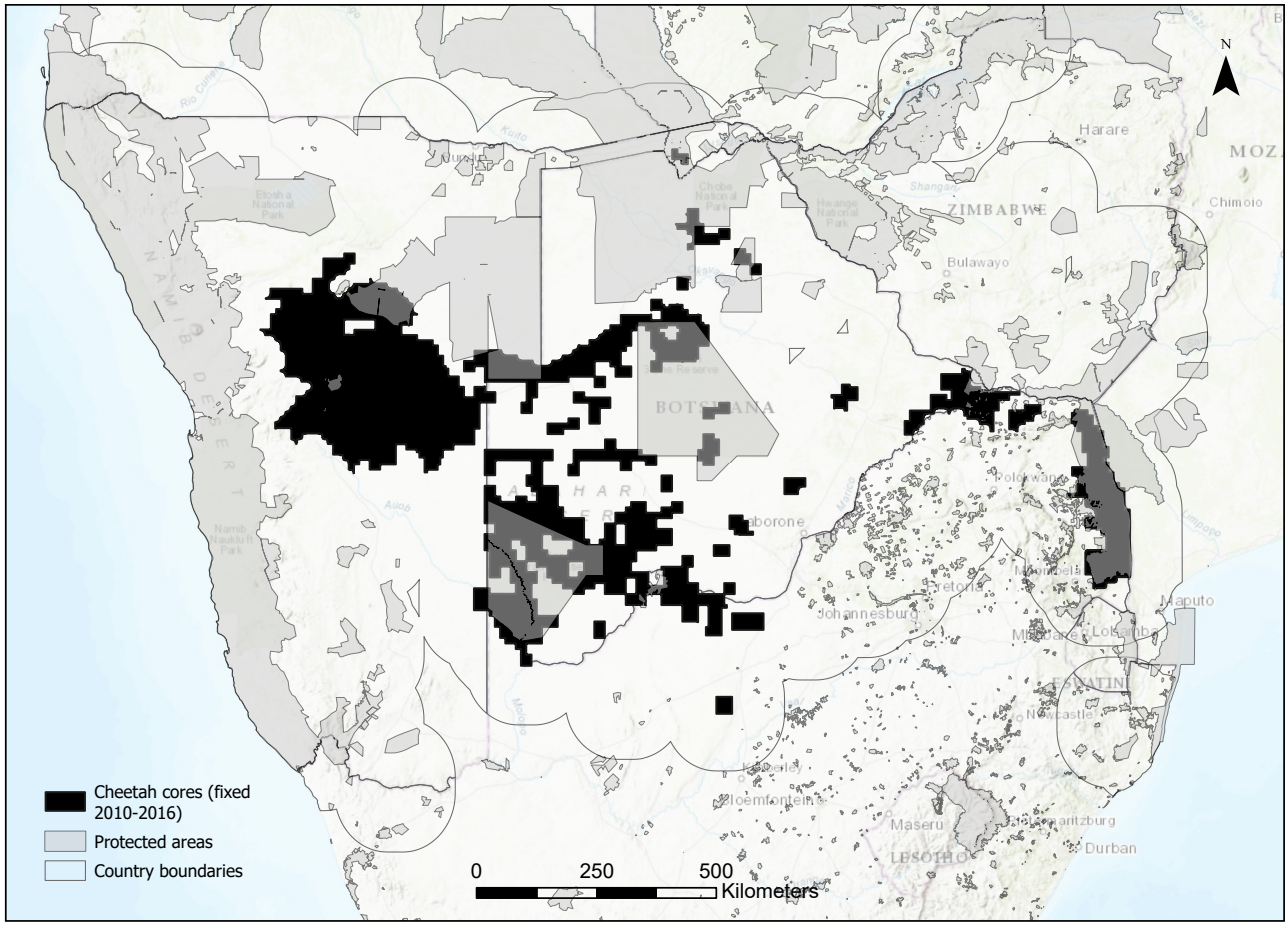

Fig. 1 Study extent, fixed historical range cores, and core groups used to define the modeled connectivity network. Core groups are an analysis construct and do not imply separate populations.

Table 1 Input layers, their data sources, how each was treated through time, and whether it entered the resistance surface.

Layer	Source	Treatment	In resistance	Role
Built-up fraction	Global Human Settlement Layer (GHSL), R2023A (Pesaresi 2023)	Dynamic	Yes	Human pressure
Nighttime radiance	Visible Infrared Imaging Radiometer Suite (VIIRS) nighttime lights, VNL V2 (Elvidge et al. 2021); gas flares masked	Dynamic	Yes	Human pressure
Road density	Global Roads Inventory Project, version 4 (GRIP4) (Meijer et al. 2018)	Static	Yes	Human pressure
Livestock density	Gridded Livestock of the World, version 4 (GLW4), 2020 (Food and Agriculture Organization of the United Nations 2024; Gilbert et al. 2018); cattle, goats, and sheep combined	Static	Yes	Human pressure
Tree, non-tree, and bare cover	Moderate Resolution Imaging Spectroradiometer (MODIS), MOD44B (DiMiceli et al. 2022)	Dynamic	Yes	Vegetation
Slope	Shuttle Radar Topography Mission (SRTM) (Farr et al. 2007)	Static	Yes	Terrain
Veterinary fences	World Wide Fund for Nature (WWF) Kavango–Zambezi Transfrontier Conservation Area (KAZA) fence map (World Wide Fund for Nature 2026); Kruger National Park fence references	Static; varied between scenarios	Yes, as a multiplier	Raises resistance where present; crossable, not a full barrier
Land cover	MODIS land cover, MCD12Q1 (Friedl and Sulla-Menashe 2022)	Context only	No	Masking and description; never differenced between years
Hydrography	HydroSHEDS rivers and lakes (Lehner and Grill 2013; Lehner et al. 2008; Messager et al. 2016)	Context only	No	Description and checks; water not treated as a barrier
Protected areas	World Database on Protected Areas (WDPA), August 2026 WDPA release (UNEP-WCMC and IUCN 2026)	Overlay	No	Reporting protected-area coverage only

Note: Dynamic layers change between the 2012, 2016, 2020, and 2024 snapshots; static layers are the same in every snapshot. The product year matched to each snapshot is given in Appendix B. VNL = VIIRS nighttime lights; V2 = product version 2; MOD44B = MODIS vegetation continuous fields; MCD12Q1 = MODIS land cover type; HydroSHEDS = hydrological data and maps based on Shuttle Elevation Derivatives at multiple Scales. Component weights are given in the text.

[TODO: Table updated 6 October 2026: it now says which layers feed the resistance surface, adds land cover and protected areas, and cites every source. Product years were taken out because supplement Table B3 already lists them; the note points there. Check the roles match your Methods text. TWO SOURCES STILL UNCITED: (1) the Kruger fence references, which came from the SANParks park boundary plus the Kruger management plan and elephant management plan; cite those plans if you describe the Kruger fence treatment in the text. (2) The gas-flare mask, built from World Bank Global Gas Flaring Reduction flare locations for 2012–2025; cite the World Bank dataset if you mention the masking in Methods.]

ysis included all 253 core pairs in each of the four years, for a total of 1,012 pairwise comparisons.

2.6 Network structure

Edge betweeness counts how many shortest paths between cores pass through a link. A high value edge betweeness means that the link is important for holding the larger network together. The value was calculated on the 45-link network and divided by the 253 possible core pairs. The network combined each core's three nearest neighbors with a minimum spanning tree, which is the smallest set of links needed to keep the core network connected. Allowing two, three, four, or five nearest neighbors created 31, 45, 61, or 77 links, and the highest-ranked links changed. Link 17–24 was the only bridge under the three-neighbor choice, which means that removing it would disconnect the network.

2.7 Temporal comparison

Temporally, I compared the same core pairs, model assumptions, and 1-km grid in 2012, 2016, 2020, and 2024. This kept the comparison consistent so that the modeled differences came from the input layers that changed through time. Absolute change was defined as the later value minus the earlier value. Percent change was defined as that difference divided by the earlier value, multiplied by 100.

2.8 Protected-area coverage

[TODO: PLAIN-LANGUAGE WRITING GUIDE. Question: what share of the modeled connecting landscape outside the cores falls inside protected areas? Data preparation: use the August 2026 WDPA release (UNEP-WCMC and IUCN 2026). Keep terrestrial and coastal protected-area polygons, remove proposed areas, points, and marine-only areas, and merge overlaps. This left 2,301 records in the study extent. Comparison area: 31.26 percent of the usable outside-core area was protected. Cells with zero current were excluded because they could never enter a high-current group. Why no p-value: the answer changed from significant to not significant when the assumed spatial block size changed. Report the descriptive percentages and note that the same general pattern appeared in all four years and under several core-buffer sizes.]

2.9 Stability and uncertainty

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What "stable" means here: a link had to pass three checks. First, both alternative weightings had to keep at least 80 percent of the route within 5 km of the main route; 34 of 45 passed. Second, the fence scenarios could not substantially change the link; this removed two more from the group. Third, the direction of change had to stay the same under all vegetation weightings. The two links that failed this last check had already been removed. The final groups were 32 main-priority links and 13 uncertain links. Important distinction: the 80-percent-within-5-km rule is not the cell-by-cell agreement test proposed in the original protocol, even though both use the number 80 percent. What was not completed: alternative core definitions, 2-km and 5-km grids, and a cell-by-cell agreement map. Separate location uncertainty from cost uncertainty: 49 of 180 route-and-year comparisons had less than 80 percent overlap within 1 km, including 21 main-priority links. Link 17–24 had no overlap in 2020. This does not mean the connection was unimportant; it means the exact mapped route was uncertain.]

2.10 Independent plausibility assessment

[TODO: PLAIN-LANGUAGE WRITING GUIDE. UPDATED 5 OCTOBER 2026: THE PLANNED CHECK IS NOW DONE.
Never call this "validation." No resistance value was fitted to these tracks, so it is a plausibility check: does the model's picture of easy and hard land line up with where real cheetahs spent their time?

DATA: tracking data from nine free-roaming cheetahs (seven males, two females) collared near Thabazimbi, Limpopo, South Africa, from September 2003 to November 2008 by the Endangered Wildlife Trust and the University of Pretoria. Cite the dataset parencitemarnewick2017ewtdata and the paper that describes the collaring parencitemarnewick2017tracking. License CC-BY 4.0, so citing is all that is required. It is independent of your cores, and you can say why in one sentence: these collar locations are not in the Dryad range archive the cores were built from. I checked. The archive's 458 Endangered Wildlife Trust rows are 2014–2015 tourist sightings, not these collars, and only four archive records of any kind fall inside the area these cheetahs used (three farmer questionnaires and one sighting, 2012–2015). That also explains why no core sits in this area.

STEPS, matching protocol steps 23.1 to 23.5:
1. Downloaded all 3,165 locations. Dropped 17 that had no full date.
2. Kept one location per animal per day (the earliest fix) so a single busy collar could not dominate. This left 1,006 locations. One collar alone had 1,954 of the raw fixes, which is why this step matters.
3. Made ten random "available" points per cheetah location (10,060 in total), spread evenly inside the smallest shape that encloses all the cheetah locations. This asks: within the landscape these animals actually used, did they favor the places the model calls easy? Repeated with that shape widened by 20 km to check the answer did not depend on where the edge was drawn.
4. Read the modeled resistance and current at every point, using the 2012 surfaces because the tracks are from 2003–2008, the closest snapshot.
5. Compared the two groups, and repeated everything on your earlier model (pressure and terrain only, no

vegetation, no fences).

ADDED AFTER THE PLAN (5 OCTOBER 2026), FOUR STRICTER FOLLOW-UPS. Say in one sentence that these go beyond the protocol and were added because steps 3 to 5 have two weak spots: they treat a thousand daily locations as independent when the real sample is nine animals, and comparing against the whole area only shows that the animals' home ground was easier, not that they chose easier ground.

6. STEP CHECK (the main tracking result now). Took every pair of locations from the same animal on back-to-back days that were at least 1 km apart (549 moves; 303 shorter moves stayed in the same 1 km cell and 145 pairs had a gap of more than a day). For each real move, made 20 moves the same cheetah could have made instead: same starting point, a step length drawn from that animal's own moves, random direction. Then asked what share of those 20 the real move beat (lower resistance, higher current), at the end point and averaged along the straight line. 0.50 means no difference. Each animal gets one score, and the 95% interval comes from resampling the nine animals 10,000 times, so the animal is the unit, not the location. This design is borrowed from step-selection studies; cite it with parencitefortin2005wolves, thurfjell2014ssf (Fortin introduced the matched "could have gone instead" design; Thurfjell is the standard review of it). Both are in the bib.

7. LAYER BREAKDOWN. Rebuilt the main 2012 surface from its parts, using the same scaling and weights as the build scripts (it matched the stored surface at 99.97% of sampled points). Scored each part on its own, then scored the main surface with each part left out. This shows which inputs carry the result. Say clearly that nothing was reweighted afterward; this is a description, not a tuning step.

8. ALL NINE SURFACES. Repeated the step check on the main surface and the eight alternatives. This tests whether the result depends on one set of assumptions. Say it was not used to choose a surface.

9. ROAD CROSSINGS. Counted how often real moves crossed a mapped GRIP4 road compared with the alternative moves. Fences could not be tested this way: no mapped fence lies within 20 km of any track.

THINGS TO STATE OPENLY: the tracks end four years before your first snapshot; nine animals in one district cannot speak for the whole network; none of the locations fell inside a core, so the current values are not inflated by the cores; the straight line between two daily locations is not the route the animal walked; and the fence part of the model is untested by these data. Scripts in the connectivity repo: `analysis/scripts/plausibility_limpopo_tracks.py` (steps 1 to 5), `plausibility_limpopo_steps.py` (step 6), and `plausibility_limpopo_followups.py` (steps 7 to 9).

THE OLD SAME-ARCHIVE CHECK: keep it, but as a second, weaker check, and say it is not independent because it uses the records that built the cores. Numbers: after thinning to one record per 10-km cell, 2010–2012 had 413 cells and 2013–2016 had 450, with an overlap score of 0.239. There are no records after 2016.

STILL NOT DONE: the central-eastern Namibia camera-trap comparison (Verschueren et al. 2024). Those data are request-only. Either request them or state in one sentence that this second planned source was not obtained.]

All inputs to the connectivity model are static. The cores are fixed from historical records, the landscape is represented by four snapshots, and the modeled paths are estimates rather than observed routes. To assess whether this static representation bears any relation to how cheetahs actually move, the model was compared with independent GPS tracking data from free-ranging cheetahs. This comparison is a plausibility check rather than a validation, because no resistance value was fitted to these data. The tracking data consist of 3,165 locations from nine free-ranging cheetahs (seven males and two females) collared near Thabazimbi, Limpopo Province, South Africa, between September 2003 and November 2008 (Marnewick et al. 2017a; Marnewick et al. 2017b). The tracked area lies inside the study extent but outside every core, so these animals were moving through the landscape between cores, which is the part of the map the model makes claims about. The data are also independent of the cores, as none of these collar locations appear in the range archive from which the cores were derived (Weise et al. 2017a).

Seventeen records without a complete date were removed, and the remaining locations were thinned to the earliest fix per animal per calendar day, leaving 1,006 locations. Locations from the same animal on consecutive days were then paired into day-to-day moves. Moves shorter than 1 km were excluded because both ends fall within the same 1 km cell, which left 549 moves with a median length of approximately 4 km. Each move was compared with 20 alternative moves the same animal could have made instead, following the logic of step-selection analysis, a method that compares observed movements with the alternatives available from the same starting point (Fortin et al. 2005; Thurfjell et al. 2014). Each alternative began at the same location as the real move, covered a distance drawn from that animal's own movements, and headed in a random direction. Because every alternative began where the real move began, the comparison reflects the animal's choice of destination rather than the general character of the area it lived in. Modeled resistance was read from the 2012 surfaces, the snapshot closest to the tracking period, at the end point of each move and averaged along the straight line between its start and end. For each move, the share of alternatives that reached higher resistance than the real move was recorded. If modeled resistance had no bearing on move-

ment, this share would be approximately 0.50. Shares were averaged within each animal so that each of the nine animals counted once, and a 95% interval was derived by resampling the animals 10,000 times.

[TODO: Co-written 5 October 2026 from your answers. Still to add, one or two sentences each: the broader comparison against random points across the whole tracked area (steps 3 to 5 in the guide above), the layer breakdown, the repeat on all nine surfaces, and the road-crossing count.]

2.11 Monitoring prioritisation

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What was planned: combine six measurements into one score for every 10-km cell and test three sets of weights. What was built instead: a link-level table that keeps each kind of evidence separate. Its columns describe how important the link is to the network, how its modeled cost changed, how stable its mapped route was, whether fences affected it, and how much of it is protected. Why this is more honest: combining everything into one score would require deciding, for example, how much route instability is equal to a change in protection. No evidence-based exchange rate was available. State clearly that no combined monitoring score was calculated.] There is a cheeta [TODO: Unfinished sentence fragment left here; finish or delete it.]

2.12 Software and reproducibility

[TODO: PLAIN-LANGUAGE WRITING GUIDE: list the software without explaining every technical setting in the main text. Verified versions are ArcGIS Pro 3.7.2 (build 1901), ArcGIS Python 3.13.13, Julia 1.12.7, and Circuitscape.jl 5.17.1; point readers to the full software table in Appendix B. Google Earth Engine supplied and aligned the satellite data. Most ArcGIS processing used Python scripts rather than ModelBuilder. Add the final archive link and state which source datasets cannot legally be redistributed.

AI DISCLOSURE, REQUIRED BY BIODIVERSITY AND CONSERVATION. The journal says use of a large language model "should be properly documented in the Methods section," unless it was only used for copy editing (grammar, spelling, readability). You used an AI assistant for more than that: writing and debugging analysis scripts, building the tracking-check scripts, checking numbers against the source files, and drafting writing guides and some text that you then revised. Add one or two honest sentences here, for example: "An AI assistant (Claude, Anthropic) was used to help write and debug analysis scripts and to draft text, which the author reviewed, revised, and verified against the source outputs. The author takes full responsibility for the content." AI tools cannot be listed as an author. The capstone may have its own rule on AI use, so check that with your advisor too.]

3 Results

3.1 Connectivity change, 2012–2024

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What happened: I compared all 253 possible pairs of cores. Overall, the modeled cost of moving between cores changed only a little from 2012 to 2024. The middle change was -1.224 percent. A negative value means the model became slightly easier to move through; it does not mean that cheetahs actually moved more. Of the 253 pairs, 92 became harder and 161 became easier. The largest increase was +13.1 percent for the link between cores 33 and 38. What the map adds: outside the cores, the places with the highest modeled current were mostly in the same locations through time. Their overlap scores were 0.91 to 0.93, where 1 would mean perfect overlap. What not to claim: do not call this an overall connectivity loss, observed movement, or a country-by-country result. The analysis did not assign each core pair to a country. For H1, report that 18 of the 45 selected links became harder and 27 became easier, but that change was not meaningfully related to baseline betweenness ($\rho \approx 0.01$). Therefore H1 was mixed rather than supported. Use Fig. 3. When you point to a figure in your own sentences, type `\figref{fig:change}` rather than "Figure" by hand: it prints "Figure" in the capstone and "Fig." in the journal layouts, which is how Springer writes it.
H1, TWO MORE NUMBERS YOU CAN USE (checked 5 October 2026; same answer, stated the way the protocol worded H1). The $\rho \approx 0.01$ above mixes the links that got harder with the ones that got easier. Asking the protocol's question directly: (1) Did the 18 harder links have higher betweenness than the 27 easier ones? Slightly, but not meaningfully (Mann–Whitney one-sided $p = 0.36$). (2) Among the 18 harder links, did the bigger losses fall on the more important links? No; if anything the reverse ($\rho = -0.22$, not significant). One thing that DID match: the losses were fairly concentrated, with the 5 hardest-hit of the 18 links carrying 46 percent of the total increase. So H1's "minority of links" half holds, and its "high-betweenness links" half does not.
FIGURES FOR THIS SUBSECTION (moved here 6 October 2026): the 2024 current map comes first, then the change map. Mention both in your sentences, the 2024 map first, e.g. "(`\figref{fig:current2024}`)" where you describe where current concentrated, and "(`\figref{fig:change}`)" where you describe how it changed.]

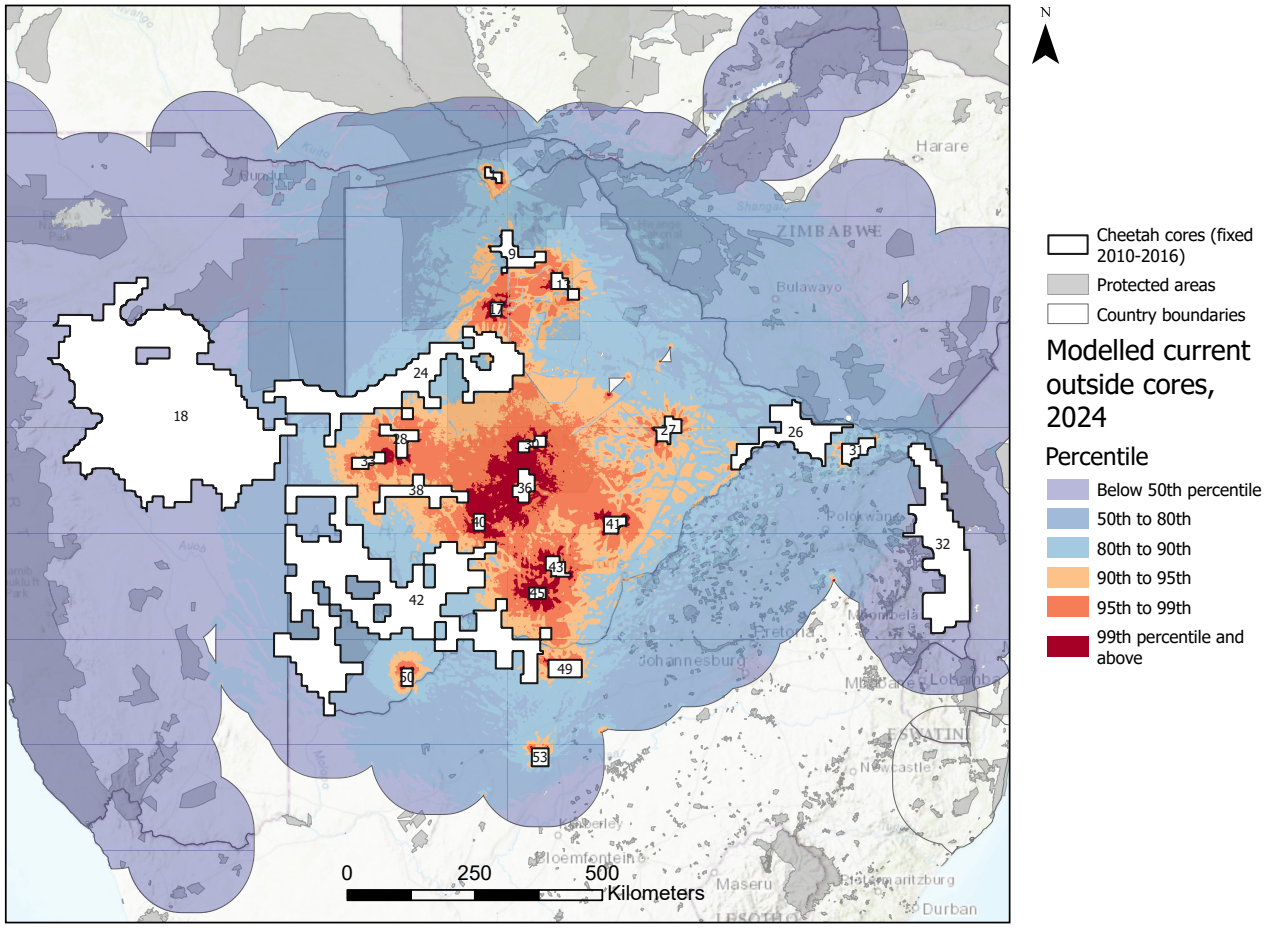

Fig. 2 Modeled current concentration for 2024, with the fixed core network shown for orientation.

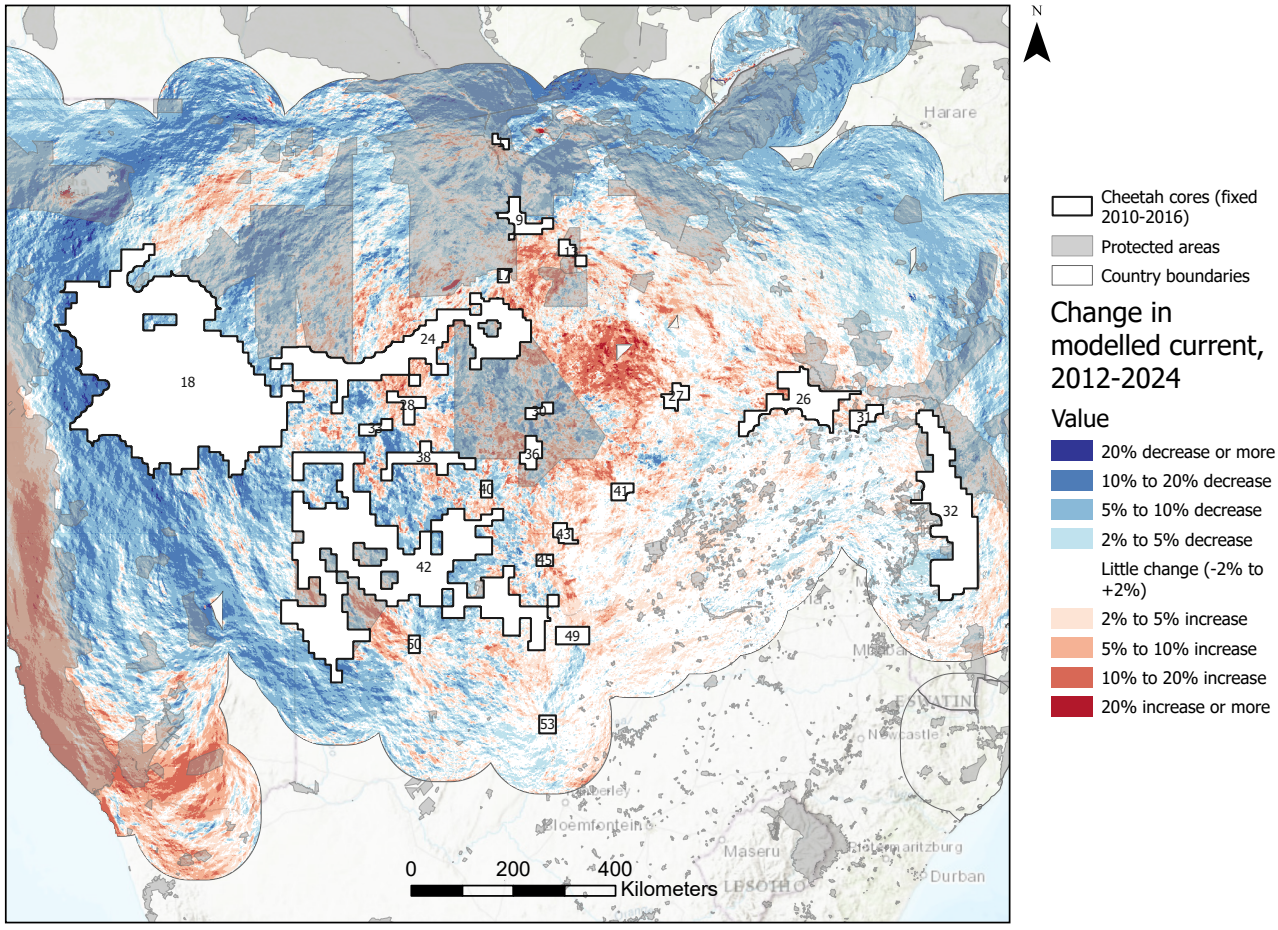

Fig. 3 Modeled current change between 2012 and 2024.

3.2 Priority links and sensitivity checks

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What happened: 32 of the 45 selected links passed the tested sensitivity screens and remained in the main priority group. Network importance was ranked separately using betweenness. The other 13 links were kept as uncertain links where more field information would be useful. The weighting test kept 34 links, and the fence test removed two more. The vegetation test did not remove any additional links from the remaining group. What it means: the main group is not a small handful of links; it contains most of the network. That only partly matches the original expectation that priorities would be limited to a small subset. What not to claim: these links are modeled connections between cores, not confirmed animal travel routes. Use Table 2. FIGURE AND TABLE FOR THIS SUBSECTION: mention the link map first, e.g. "(`\figref{fig:networkcontext}`)" where you introduce the 45 links, then the table, "(Table~`\ref{tab:toplinks}`)" where you name the highest-ranked links.]

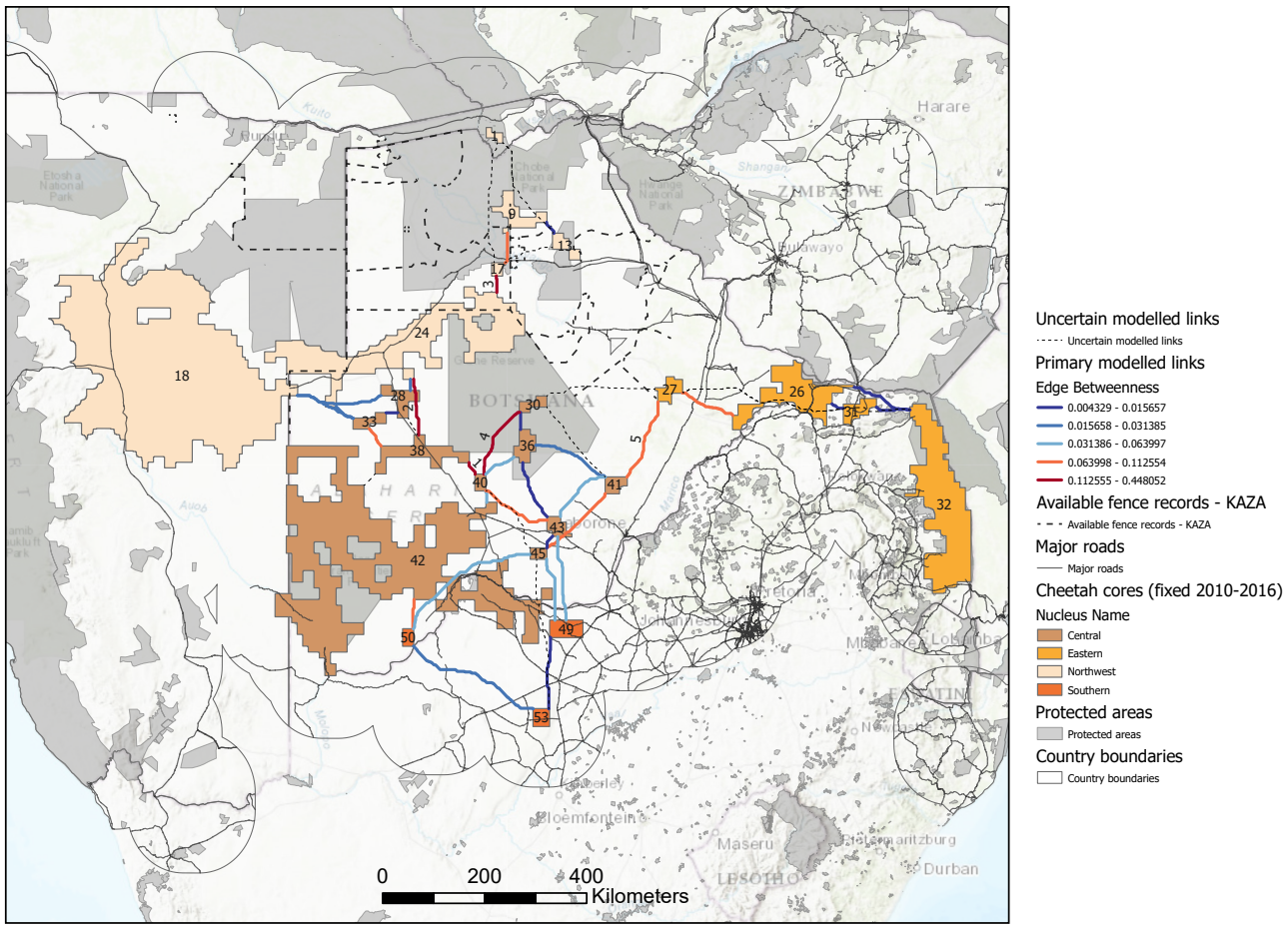

Fig. 4 Modeled core-to-core links with transportation and available fence context. Lines represent network links used to define least-cost analyses; they are not observed animal paths.

Table 2 Ten highest-ranked modeled priority links shown from the 32-link primary-priority set.

From	To	Rank	Betweenness	Type	Distance (km)	Change (%)
38	40	1	0.409	KNN+MST	120.6	+4.78
24	38	2	0.316	KNN	202.3	+7.44
17	24	3	0.300	KNN+MST	208.0	-5.78
30	40	4	0.238	KNN	169.2	+3.55
27	41	5	0.103	KNN+MST	196.8	-6.45
41	45	6	0.090	KNN	183.6	-5.50
33	38	7	0.079	KNN	108.1	+13.12
9	17	8	0.079	KNN+MST	109.0	-5.26
26	27	9	0.079	KNN+MST	221.8	-5.18
40	43	10	0.077	KNN	159.6	-1.56

Notes: The table shows the ten highest-ranked members of the 32-link primary-priority set; links are modeled structural network relationships, not observed animal routes. KNN = k-nearest-neighbor link; MST = minimum spanning tree link. "KNN+MST" identifies links retained by both network-construction rules.

[TODO: PLAIN-LANGUAGE WRITING GUIDE FOR THE SENSITIVITY TESTS. Main lesson: the total modeled cost of a connection was usually fairly stable, but the exact line taken by the least-cost path was less stable. Of 180 route-and-year comparisons, 49 had less than 80 percent overlap within 1 km. This affected 21 of the 32 priority links. The route between cores 17 and 24 had no overlap in 2020. Vegetation assumptions: 43 of 45 links kept the same direction of change under all three vegetation weightings. However, the size of the change grew as vegetation was given more weight: the middle absolute change increased from about 3.6 to 9.3 percent. Network-design assumption: changing how many neighbors each core could connect to changed the number of links and changed which links ranked highest. A bridge is a link whose removal splits the network into separate pieces; link 17–24 was the only bridge when three nearest neighbors were used. What not to claim: there is no completed cell-by-cell agreement map, and no single ranking should be presented as certain. State that H3 was only partly evaluated: a stable group existed, but it contained a majority of the selected links and the full planned sensitivity grid was not completed.]

3.3 Independent tracking check

[TODO: PLAIN-LANGUAGE WRITING GUIDE FOR THE INDEPENDENT TRACKING CHECK (UPDATED 5 OCTOBER 2026 WITH THE STRICTER FOLLOW-UPS).
This now has its own subsection, because it answers a different question from the priority links; rename the heading if you like. Numbers only here; meaning goes in Discussion. Lead with the step check (Table 3), because it is the stricter test. Put the first comparison (Table 4) after it as background, or move that table to Appendix B.

1. THE MAIN RESULT (Table 3, panel A). Across 549 day-to-day moves, the place a cheetah actually went had lower modeled resistance than the moves it could have made instead. The real move beat 56 percent of its alternatives (95 percent interval 51 to 61 percent), where 50 would mean no difference, and seven of the nine animals leaned that way. Averaged along the straight line between locations the result was almost the same (55 percent).
2. CURRENT DID NOT HOLD UP. The real move beat 53 percent of alternatives on modeled current, but the interval (47 to 59) includes 50, and only five of nine animals leaned that way. Say this plainly. It also means the 65 percent for current in the first comparison mostly reflected where the cats lived, not the choices they made each day.
3. ALL NINE SURFACES (panel B). Every surface gave 55 to 56 percent, so the result does not depend on which weighting you picked. The four other fence treatments give exactly the same values as the main surface because no mapped fence lies within 20 km of any track.
4. WHICH LAYERS CARRY IT (panel C). Road density did most of the work: on its own it reached 55 percent, about as much as the full surface, and leaving it out dropped the full surface to 54 percent with an interval that just touches 50. Livestock, slope, and night lights each added a small lean (51 to 52 percent). Built-up area did nothing here because there is almost none in this district. Vegetation on its own leaned slightly the wrong way (48 percent, interval 43 to 52, which includes 50). Report that honestly; do not hide it.
5. ROAD CROSSINGS (panel D). 26 percent of real moves crossed a mapped road, against 35 percent of the alternatives; eight of nine animals crossed less than expected. Major roads were crossed too rarely by anyone (1 against 2 percent) to say much.
6. THE FIRST COMPARISON (Table 4), as background: against random points across the whole area, cheetah locations had lower resistance 60 percent of the time (7 of 9 animals) and higher current 65 percent of the time (6 of 9). Widening the area by 20 km and using the earlier model gave the same picture.

HOW TO DESCRIBE THE STRENGTH: consistent in direction, modest in size, and mostly carried by roads. 56 percent is a real lean, not a strong one. Say "consistent with" the model, not "confirmed." Report the intervals and animal counts, not p-values.

WHAT NOT TO CLAIM: that the model is validated; that it works across the whole region (one district in South Africa); that it says anything about 2020 or 2024 (the tracks end in 2008); or that fences were tested (they were not).]

[TODO: PLAIN-LANGUAGE WRITING GUIDE FOR THE SAME-ARCHIVE OCCURRENCE CHECK. Keep this, but present it after the tracking check as the weaker of the two, because it is not independent. What happened: after thinning records to one 10-km cell, the early period had 413 occupied cells and the later period had 450. Their overlap score was 0.239, showing that the two sets were quite different. Later records occurred in slightly higher modeled resistance, and a larger share fell inside the fixed cores. Almost no records were close to the priority paths: zero early cells and three later cells. Why this is a weak check: the same occurrence archive helped create the cores, so this is not an independent test. There are also no occurrence records after 2016, so it cannot check the 2020 or 2024 results. Report it honestly as a limited comparison, not as validation.]

[TODO: Protocol reporting only: the planned cell-level monitoring index was not built and H4 was not tested. Explain this once in Appendix A, not as a repeated priority-link result.]

3.4 Protected-area coverage of priority connectivity

[TODO: PLAIN-LANGUAGE WRITING GUIDE. Why the cores were removed: the model forces current into and out of the cores, so the cores automatically receive very high current. Including them would make protec-

Table 3 [TODO: Final caption, in your own words: day-to-day moves of nine free-roaming cheetahs (Limpopo, 2003–2008) compared with moves of the same length the animals could have made instead, on the 2012 surfaces. Panels: main surface, all nine surfaces, single input layers, road crossings.]

Comparison	Share of alternatives the real move beat	95% interval (resampling animals)	Animals above 0.50
A. Main surface, 549 day-to-day moves, 20 alternatives each			
Resistance at the end point (lower expected)	0.56	0.51–0.61	7 of 9
Resistance along the straight-line path	0.55	0.51–0.59	7 of 9
Current at the end point (higher expected)	0.53	0.47–0.59	5 of 9
Current along the straight-line path	0.52	0.47–0.58	5 of 9
B. All nine surfaces, resistance at the end point			
Main surface	0.56	0.51–0.61	7 of 9
Equal human-pressure weights	0.56	0.51–0.61	7 of 9
Infrastructure emphasis	0.56	0.50–0.61	6 of 9
Vegetation weight 10%	0.56	0.50–0.61	7 of 9
Vegetation weight 40%	0.55	0.51–0.60	6 of 9
Four other fence treatments[a]	0.56	0.51–0.61	7 of 9
C. Single input layers, resistance at the end point			
Road density	0.55	0.50–0.62	6 of 9
Livestock	0.52	0.49–0.56	5 of 9
Terrain (slope)	0.52	0.50–0.54	4 of 9
Night lights	0.51	0.51–0.52	8 of 9
Built-up area[b]	0.50	0.50–0.50	3 of 9
Vegetation	0.48	0.43–0.52	3 of 9
Main surface without road density	0.54	0.50–0.57	7 of 9
D. Road crossings along the straight-line path (fewer expected)			
Any mapped road (26% of real moves crossed, 35% of alternatives)	0.54	0.52–0.57	8 of 9
Major roads only (1% against 2%)	0.51	0.50–0.51	5 of 9

[TODO: Final note, in your own words. Each real day-to-day move (at least 1 km) is compared with 20 moves the same cheetah could have made instead: same start, a step length drawn from that animal's own moves, random direction. The score is the share of those alternatives the real move beat; 0.50 means no difference. Each animal counts once, and the 95% interval comes from resampling the nine animals. [a]No mapped fence lies within 20 km of any track, so all five fence treatments give identical values here; these data cannot test the fence part of the model. [b]Built-up area barely varies in this district, so it cannot separate real and alternative moves.]

Table 4 [TODO: Final caption, in your own words: modeled resistance and current at daily locations of nine free-roaming cheetahs (Limpopo, 2003–2008) compared with random available points from the same landscape, on the 2012 surfaces. Call it the broader, less strict comparison.]

Surface	Measure	Cheetah fixes (median)	Available (median)	Expected direction (share)	20-km buffer (share)	Animals (of 9)
Final model (vegetation-balanced, documented fences)	Resistance (lower expected)	2.23	2.45	0.60	0.60	7
	Current (higher expected)	0.156	0.145	0.65	0.62	6
Earlier model (pressure and terrain only, no fences)	Resistance (lower expected)	2.03	2.34	0.61	0.61	–
	Current (higher expected)	0.143	0.128	0.65	0.63	–

[TODO: Final note, in your own words: the comparison uses 1,006 daily locations from nine free-roaming cheetahs (2003–2008) against 10,060 random points from the same landscape, on the 2012 surfaces. "Expected direction (share)" is the chance that a random cheetah location had lower resistance (or higher current) than a random available point; 0.50 means no difference. "20-km buffer" repeats this with a wider available area. "Animals" counts how many of the nine had a median in the expected direction. No location fell inside a core. Add one sentence saying this is the broader comparison, and that in the stricter step check (Table 3) the resistance lean held but the current lean did not.]

tion look better without telling us how well the land between cores is protected. What happened outside the cores: 31.26 percent of the comparison area is protected. Only 23.27 percent of the area above the 80th current percentile was protected in 2024. Coverage rose to 25.54 percent above the 90th percentile, 33.45 percent above the 95th, and 44.63 percent above the 99th. In simple terms, the very strongest concentrations often reached protected areas, while the broader connecting landscape around them was less protected. What not to claim: do not report a significance p-value because it changed when the size of the spatial blocks changed. Do not call modeled current observed movement. Use Table 5. State that H2 was not supported under its full rule because the direction changed across current thresholds and the significance result was not stable.]

Table 5 Protected-area coverage of modeled high-current classes.

Year	Current percentile	Cells	Protected (%)	Background (%)	Difference (pp)
2012	80	418,295	24.23	31.26	-7.03
2012	90	209,148	26.31	31.26	-4.95
2012	95	104,574	35.57	31.26	+4.31
2012	99	20,915	44.17	31.26	+12.92
2024	80	418,292	23.27	31.26	-7.99
2024	90	209,146	25.54	31.26	-5.72
2024	95	104,573	33.45	31.26	+2.19
2024	99	20,915	44.63	31.26	+13.37

[TODO: Final note: values use the outside-core, positive-current domain. Percentage-point differences are relative to the matched background. Do not report a permutation p-value because inference was sensitive to the assumed spatial block size.]

4 Discussion

4.1 Connectivity change and comparison with previous work

[TODO: PLAIN-LANGUAGE GUIDE: begin with the simplest overall result: the modeled ease of moving between cores changed only slightly overall, but some individual links changed more than others. Immediately state that the model describes landscape structure and does not show where cheetahs actually moved.]

[TODO: PLAIN-LANGUAGE GUIDE: describe where modeled connections became easier or harder and which landscape changes occurred in the same places. Say "occurred in the same places," not "caused." The model found mixed change rather than a broad decline. Do not describe connectivity by country because core pairs were not assigned to individual countries.]

[TODO: PLAIN-LANGUAGE GUIDE: compare the general mapping approach with the elephant and multi-species studies (Brennan et al. 2020; Naidoo et al. 2024) and with the fence study (Osipova et al. 2018). Explain that similar mapping tools can be used for different species, but the numerical resistance values cannot simply be copied because each species responds differently to roads, vegetation, fences, and people.

FENCES, ADDED 6 OCTOBER 2026: cite (Rosen et al. 2026) here. What that paper found: in northern Botswana many veterinary fences are already in disrepair from elephant damage, though some sections still block important wildlife movement. Their disease-risk assessment of sections of the Northern Buffalo, Zambezi Border, and Western Border fences found that removing those sections would not raise the risk of foot-and-mouth disease and two other livestock diseases, and that with measures such as better herding the risk could be lower than now. Stakeholders agreed to start consultations on removing low-risk sections. Two ways it connects to your paper, in your own words: (1) it supports treating fences as raising resistance rather than as complete barriers, which is how your model handles them, because real fences are often broken; and (2) it makes your fence scenarios relevant to a live policy question, since the links that changed under the fence tests are the kind of places where opening fence sections would matter most. Keep it to one or two sentences, and do not claim your model shows what removing a fence would do for cheetahs, because no fence-removal scenario was run.]

4.2 Protected areas and conservation context

[TODO: PLAIN-LANGUAGE GUIDE: the strongest modeled concentrations often pass through protected areas, but the wider land connecting those areas is less protected. Explain that conservation therefore also depends on farms, communal lands, and other land outside formal parks. This does not prove that protected areas failed. Also note that parks are often located on land with lower agricultural value, which may partly shape the pattern.]

4.3 Priority links and field-survey needs

[TODO: PLAIN-LANGUAGE GUIDE: links that stayed important under several assumptions are the strongest candidates for conservation attention. Links that changed substantially between assumptions are useful places for field surveys because better local data could change the result. Agreement between scenarios is not complete proof because the scenarios share many of the same input layers.]

[TODO: PLAIN-LANGUAGE GUIDE: there is no completed score that ranks individual survey cells. Instead, explain how an organization could use the 13 uncertain links to choose general survey areas, then adjust the order based on access, cost, safety, and local knowledge. Link 17–24 is especially useful for checking because it is the only link holding two parts of the three-neighbor network together, yet its mapped route was highly unstable and had no overlap in 2020.]

4.4 How far the tracking check supports the model

[TODO: PLAIN-LANGUAGE GUIDE (UPDATED 5 OCTOBER 2026). One or two short paragraphs.
WHAT IT SHOWS: in one district, real free-ranging cheetahs moving from one day to the next tended to pick ground the model rates as easier than the ground they could have reached instead. This held for seven of nine animals and on all nine surfaces. A useful framing: the model passed a check it could have failed, because if the cats had picked higher-resistance ground, the result would have pointed the other way.
HOW BIG: modest. The real move beat about 56 percent of its alternatives, against 50 by chance. Call it a lean, not a match.
WHAT DRIVES IT: mostly roads. Road density alone did about as well as the full surface, and the cats crossed mapped roads less often than their alternative moves did. This fits work showing cheetahs avoid roads and busy areas (cite parenciteklaassen2018edge: GPS-collared cheetahs in the Maasai Mara avoided areas near roads and lodges, especially in the busy wet season; your result is the same direction in a very different, ranchland setting). Be honest about the other side: vegetation on its own leaned slightly the wrong way, and modeled current showed no clear lean. Two possible reasons for the vegetation result, say them as possibilities, not findings: cheetahs in this ranchland may use denser bush for cover and hunting, and the tracks are older than the 2012 vegetation layer.
WHAT IT DOES NOT SHOW: that the model is right elsewhere (nine animals, one part of South Africa); anything about change over time (the tracks end in 2008); or anything about fences (none are mapped near the tracks, and the private game fences in this area are not in the model at all). Do not change any weight because of this check, and say you did not; using the tracks to tune the model would make the check no longer independent.
CLOSE WITH: the planned Namibia camera-trap comparison was not done, and independent movement data from fenced and more developed areas would be the most useful next test.]

4.5 Limitations

[TODO: SENTENCE 1: Say that the model shows where movement may be easier or harder, but it does not prove where cheetahs actually traveled. SENTENCE 2: Say that the cores and several landscape layers were kept constant through time, allowing a fair comparison but limiting what could change between years. SENTENCE 3 (CORRECTED 5 OCTOBER 2026): say that the resistance values were set by you, guided by published studies, and were not fitted to cheetah movement data. The old prompt said they "came from published research," which overstates it, because the weights, the slope ramp, and the fence multiplier were your choices. In the same sentence, say that rainfall was not checked, so some vegetation change may reflect wet or dry years rather than land use, and that source years did not always match the snapshot years. SENTENCE 4: Say that the mapped paths are broad planning areas rather than exact travel routes, that the only independent check came from nine animals in one district before 2012, and that the results should be tested before use at another scale or in another region. Add one clause that this check could not test the fence part of the model, because no mapped fence lies near the tracks, and that the lean it found came mostly from roads. Optional clause that heads off an obvious reviewer question: longer-running tracking datasets exist for other parts of the network (Okavango Delta, central Namibia), but they are held privately and were not requested within the project.
LAYOUT NOTE: in the capstone this is its own section. In the journal layouts it prints automatically as the last subsection of the Discussion, because Springer research articles close with the Discussion. Write it the same way either way.]

5 Conclusion

[TODO: PLAIN-LANGUAGE GUIDE: use only three or four sentences. First, say that modeled connectivity changed only slightly overall and that individual links changed in both directions. Second, say that the strongest concentrations often reached protected areas but the wider connecting landscape was less protected. Third, state that 32 links were retained as main priorities and 13 were identified as places needing more field information. End by reminding the reader that these are modeled landscape connections, not observed cheetah routes. Do not add new results here.]

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Statements and Declarations

Data availability

[TODO: PLAIN-LANGUAGE GUIDE: add the final repository or archive link and DOI. State that the scripts and outputs created by the project are available there. Explain that raw protected-area data must be downloaded from Protected Planet under its license and that cheetah occurrence locations can only be shared at the detail allowed by their source. FOR BIODIVERSITY AND CONSERVATION (Springer research data policy): the analysis repository is private right now, so before submitting either make it public or archive a copy on Zenodo, which gives it a DOI to cite here. Springer's example wording: "The datasets generated during and/or analysed during the current study are available in the [name] repository, [link]."]

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[TODO: PLAIN-LANGUAGE GUIDE: list the department, advisor, data providers, and any organization or person who reviewed the project or helped interpret priorities. Data providers to thank by name: the Endangered Wildlife Trust and University of Pretoria (Limpopo tracking data, via GBIF), Weise and coauthors (range archive), and UNEP-WCMC (protected areas). If your advisor shaped the design or analysis rather than only grading it, offer co-authorship instead of a thank-you here.]

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[TODO: BIODIVERSITY AND CONSERVATION REQUIRES THIS STATEMENT; submissions without it are returned as incomplete. If nobody paid for the work, Springer's own wording is: "The authors did not receive support from any organization for the submitted work." If a scholarship, department, or grant helped, name it and give any grant number.]

Competing interests

The author has no relevant financial or non-financial interests to disclose.

Ethics approval

Not applicable. The study used existing, publicly available data and involved no handling of animals or human participants.

Author contributions

[TODO: SPRINGER REQUIRES THIS. Single author: "Liam Crettol designed the study, carried out the analysis, and wrote the manuscript." If your advisor becomes a co-author, say what each person did, for example who designed the study, who ran the analysis, who wrote the first draft, and who revised it. All authors must read and approve the final version, and the statement should say so.]

A Analysis protocol as predeclared

[TODO: FOR BIODIVERSITY AND CONSERVATION: this appendix is not part of the submitted manuscript. It becomes a separate file, Online Resource 1, uploaded as `ESM_1.pdf`. Springer asks that every supplementary file include, at the top, the article title, journal name, author names, and the corresponding author's affiliation and email, plus a short caption describing the file. In the main text it is referred to as "Online Resource 1"; typing `\suppA` prints that automatically in the submission layout and "Appendix A" in the capstone.] [TODO: PLAIN-LANGUAGE GUIDE FOR THE PROTOCOL APPENDIX. The saved protocol is tagged `protocol-v1.0` in the `cheetah-connectivity` repository at commit `d7de8675a074dfbffc0c7c856bb5645e1e13867d`, dated 18 August 2026. However, the file itself still says version 0.1 draft. Do not change the old file. Explain that the Git tag marked the saved version even though its internal label was not updated. Then list, in simple terms, what changed between the plan and the final analysis: the core definition, sensitivity tests, definition of a stable link, scenario names, network measurements, protected-area comparison, monitoring index that was not built, and 2030 scenarios that were added later. Include the objective and hypothesis outcomes: O1 was completed; O2 was narrowed; O3 was replaced rather than completed; H1 was mixed; H2 did not pass its full rule; H3 was only partly tested and did not match the "limited subset" prediction; and H4 was not tested.]

[TODO: ADDED 5 OCTOBER 2026: FOUR MORE ITEMS FOR THE CHANGE LIST, plus one correction.
1. A MAIN SURFACE WAS CHOSEN. The protocol said no resistance scenario would be treated as the main answer. The paper uses the vegetation-balanced surface with documented fences as the main one and reports the rest as sensitivity. Say this was a change and give your reason in one sentence.
2. WHEN THE WEIGHTS WERE SET. Say plainly whether the final weights (60/20/20 and 30/30/30/10) were fixed before or after you had looked at resistance maps or results. This is the first thing a reader checks in a predeclared study. Either answer is acceptable if it is stated honestly; if any weight was adjusted after seeing outputs, say which one and why.
3. PC/dPC WAS DROPPED. The protocol planned probability of connectivity at 50, 100, 200, and 400 km. Only betweenness was calculated. Give the real reason (see the TODO in the Network structure subsection).
4. ONE CORE DEFINITION, NOT THREE. The protocol planned three core definitions with minimum areas of 2,000, 1,000, and 500 km^2. The analysis used one density-based definition. List it here if it is not already covered by "core definition" above.
CORRECTION: the old list included "occurrence check" as a change. That is no longer true. The planned independent tracking check (protocol steps 23.1–23.5) was

completed using the Limpopo data. What is still different from the plan: the Namibia camera-trap comparison was not done, and two kinds of analysis were added. First, a second, same-archive occurrence check. Second, four stricter follow-ups to the tracking check (a step check comparing each real move with moves the animal could have made instead, a breakdown by input layer, a repeat on all nine surfaces, and a road-crossing count). List these as additions, say they were added after the planned check was run, and say that none of them changed any weight or the choice of main surface.]

B Supplementary material

[TODO: FOR BIODIVERSITY AND CONSERVATION: this becomes Online Resource 2, uploaded as `ESM_2.pdf`, with the same header details as Online Resource 1 (article title, journal name, author names, corresponding author's affiliation and email). Number its tables S1, S2, and so on, and refer to them in the main text as, for example, "Online Resource 2, Table S4"; `\suppB` prints the "Online Resource 2" part.] [TODO: PLAIN-LANGUAGE GUIDE: the tables below preserve details that are useful for checking the analysis but are too long for the main paper. They cover which source year matched each modeled year, strongly related predictors, overlap of high-current areas, software versions, vegetation assumptions, route stability, and the 2030 scenarios. Do not add tables for different map resolutions or a cell-level monitoring score because those analyses were not completed.]

[TODO: The results overview table above was moved here from the main Results on 6 October 2026. It summarizes results that already have their own figures and tables, so it works better as a reference in the supplement. If you want it back in the main paper, it would fit best at the start of the Results, mentioned in the first sentence.]

[TODO: PLAIN-LANGUAGE DATA CHECK: every value in these tables should come from a CSV in `tables/source_data/`. Never copy a number from a screenshot or estimate it from a map legend.]

Table B1 Overview of completed modeled-connectivity results.

Metric	Value	Interpretation
Final modeled-current snapshots	4 (2012, 2016, 2020, 2024)	253/253 focal-region pairs connected each year
Fixed density cores	23	Historical baseline, fixed across snapshots
Selected neighboring links	45	Core-pair comparison set
Primary-priority links	32	Stable under weight/fence screens
All-pair median effective-resistance change	-1.224%	2012–2024
Selected-link median change	-1.557%	2012–2024
Primary-priority median change	-2.618%	2012–2024
Largest all-pair increase	+13.119%	Link 33–38
Largest all-pair decrease	-8.243%	Link 43–45

Table B2 [TODO: Final caption: software environment used for the completed analysis.]

Component	Version	Verification
ArcGIS Pro	3.7.2 (build 1901)	`arcpy.GetInstallInfo()`
ArcGIS Python	3.13.13	ArcGIS Pro environment
Julia	1.12.7	`julia -version`
Circuitscape.jl	5.17.1	Julia package environment
Google Earth Engine	Cloud service	Acquisition and harmonisation

Table B3 [TODO: Final caption: temporal matching of dynamic covariates to the four analysis snapshots.]

Covariate	Source	Treatment	2012 snapshot	2016 snapshot	2020 snapshot	2024 snapshot
Built-up fraction	GHSL	Dynamic	2010 (−2)	2015 (−1)	2020 (0)	2025 (+1)
Nighttime radiance	VIIRS VNL V2	Dynamic	2013 (+1)	2016 (0)	2020 (0)	2024 (0)
Tree cover	MODIS MOD44B C6.1	Dynamic	2012 (0)	2016 (0)	2020 (0)	2024 (0)
Non-tree vegetation	MODIS MOD44B C6.1	Dynamic	2012 (0)	2016 (0)	2020 (0)	2024 (0)
Bare ground	MODIS MOD44B C6.1	Dynamic	2012 (0)	2016 (0)	2020 (0)	2024 (0)
Land cover	MODIS MCD12Q1 C6.1	Context only	2012 (0)	2016 (0)	2020 (0)	2024 (0)

[TODO: Final note: values in parentheses are source-year minus nominal-year offsets. Define GHSL, VIIRS, VNL, and MODIS here.
MISSING ROW: the source CSV has a rainfall row (CHIRPS annual precipitation) marked "not yet acquired." The table leaves it out, which hides that the rainfall check was never done. Your Limitations prompt already mentions rainfall. Pick one: add the row back and say "planned, not completed," or keep it out and make sure the Limitations sentence says plainly that rainfall was not checked. Either is fine; just do not leave it silent, because vegetation change can come from a wet or dry year rather than from land use.]

Table B4 [TODO: Final caption: predictor pairs with absolute Spearman correlation of at least 0.75.]

Predictor pair	Spearman ρ
Tree cover–bare ground	−0.877
Non-tree vegetation–bare ground	−0.836
Cattle–goats	0.774

[TODO: Final note: report only pairs with $|\rho| \geq 0.75$ from the archived 2024 predictor screen.]

Table B5 [TODO: Final caption: pairwise overlap of outside-core high-current cells among successive snapshots and endpoints.]

Earlier year	Later year	Earlier cells	Later cells	Intersection	Union	Jaccard
2012	2016	209,148	209,150	200,069	218,229	0.917
2016	2020	209,150	209,147	201,179	217,118	0.927
2020	2024	209,147	209,146	199,881	218,412	0.915
2012	2024	209,148	209,146	199,511	218,783	0.912

[TODO: Final note: cells are the outside-core top decile of positive modeled current in each snapshot.]

Table B6 [TODO: Final caption: conditional 2030 scenario changes in selected-link least-cost path cost relative to 2024.]

Scenario	Links	Median change (%)	Minimum (%)	Maximum (%)
2024 held constant (reference)	45	0.00	0.00	0.00
Half of the 2012–2024 trend extended	45	−3.36	−8.53	+6.06
Full 2012–2024 trend extended	45	−6.94	−17.12	+8.65

[TODO: Final note, in your own words. Change is relative to the 2024 surface. Only built-up area and vegetation (tree, non-tree, bare) were pushed forward along their 2012–2024 straight-line trend; nighttime lights, roads, livestock, terrain, fences, and the cores were held at 2024. These are "what if the recent trend kept going" tests, not forecasts.
WHY THE ROWS WERE RENAMED: they used to say "Low growth," "Continuation," and "High development." Those names were misleading. The "low growth" row is zero by construction because nothing changed. The "high development" row extends the whole trend, and that includes vegetation. Vegetation made links easier from 2012 to 2024, so extending it makes links easier still (−6.94%). A reader who sees "high development" expects connections to get harder, so the old label would look like a mistake.
WHAT TO SAY ABOUT IT: one sentence noting that the projected change is mostly driven by the vegetation trend, not by new development. If you want a true "more development" test, it would need built-up and lights pushed forward with vegetation held at 2024; that run does not exist yet.]

Table B7 [TODO: Final caption: sensitivity of 2012–2024 selected-link cost change to vegetation weighting.]

Vegetation weighting	Links	Median absolute change (%)
Low	45	3.57
Balanced	45	6.27
High	45	9.28

[TODO: Final note: 43 of 45 links retained the same 2012–2024 change direction under all three vegetation weightings.]

Table B8 [TODO: Final caption: route-location sensitivity under the corrected vegetation inputs.]

Link set	Links	Route-years	Route-years below 80%	Affected links
All selected links	45	180	49	29
Primary-priority links	32	128	34	21

[TODO: Final note: the threshold requires at least 80 percent reciprocal route overlap within 1 km; it is a sensitivity screen, not a biological cutoff.]

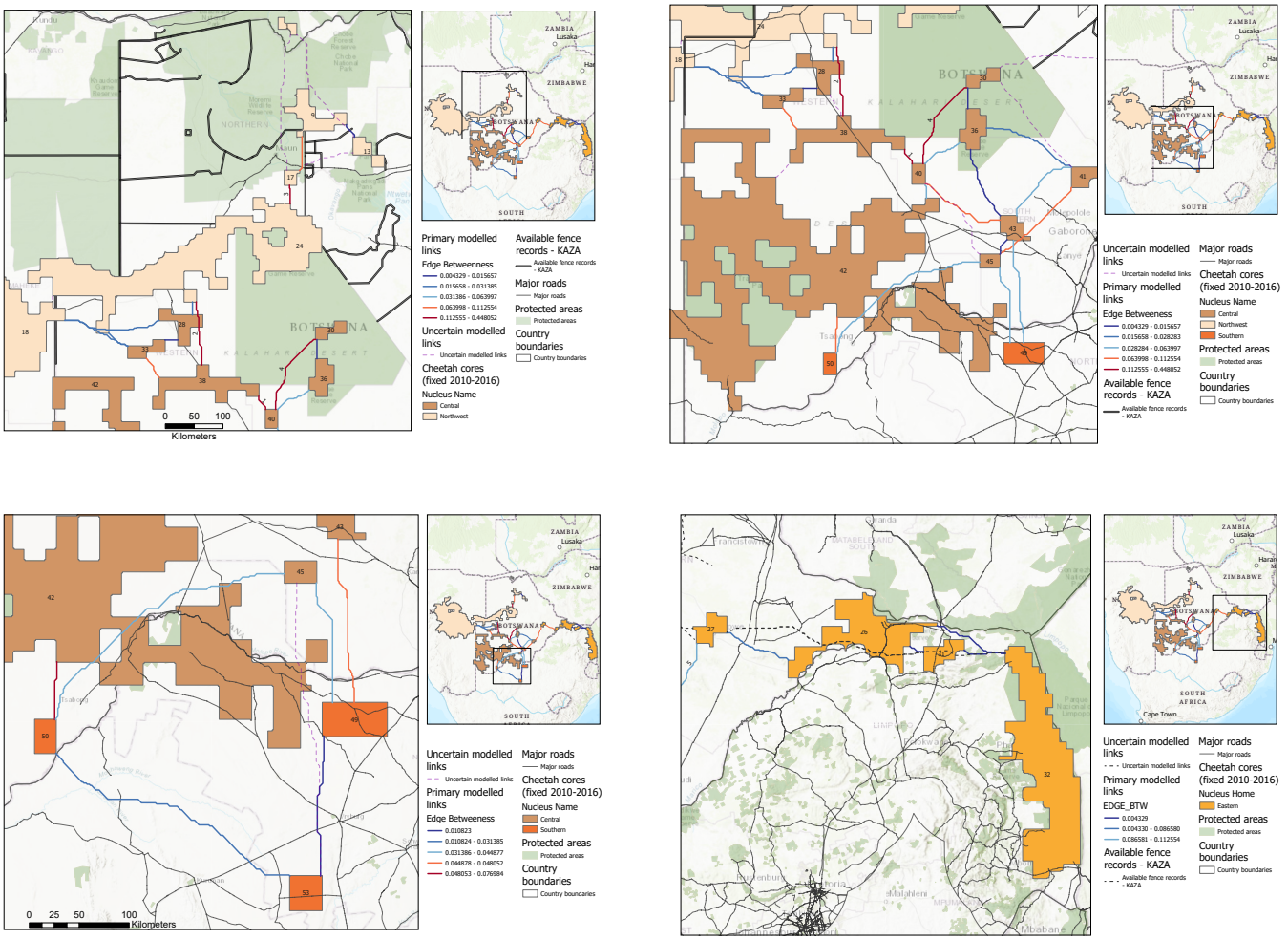

Fig. B1 Regional detail maps corresponding to the modeled-link network. These panels support visual inspection of local context and are supplementary to the main network and priority-link figures.